Finite-Group Trace Formulas:
Compact Global and Local Analogies, and a p-Adic Bridge

Corrected and expanded version of the 1996–2003 notes

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Abstract

For a finite group G, the trace of a convolution operator on the permutation representation $\mathbb{C}[\Gamma\backslash G]$ has two descriptions: one involving conjugacy classes and orbital sums, the other involving irreducible representations and their characters. This elementary identity models the Selberg trace formula for a compact quotient. The analogous bilinear identity on $\mathbb{C}[G]$ is a finite counterpart of the *compact* case of Arthur's local trace formula. An important observation is that the latter identity follows from the former by applying it to $(G \times G, \Delta G)$, where ΔG is the diagonal subgroup. We carefully fix counting-measure conventions, give proofs and explicit examples for S_3 and A_5, and explain both the value and the limits of the comparison with Arthur's global and local theories. An additional section explains, through an exact $\mathrm{GL}_2(F)$ divergence calculation, why the noncompact p-adic local formula requires truncation, Levi subgroups, and weighted distributions. Throughout, we indicate how the finite-group formulations relate to Arthur's finite example, Vign'eras's regular-bimodule method, Yang's coset-space formulas, and Terras's finite spectral analogues. A literature guide, glossary, exercises, and an account of corrections appear at the end.

Historical and editorial note. The underlying exposition began as notes written by David Joyner during a Stanford visit in summer 1996, with additions in 1997, 2000, and 2003. The present revision corrects several representation-theoretic and normalization errors, adds the diagonal-subgroup interpretation of the local formula, and supplies background and references. The present edition also adds a worked p-adic bridge. Its finite-group ideas have important precedents in Arthur [1], Vign'eras [2], and Terras [3]; Yang's later work [4] treats closely related coset-space representations. The original 1996–2003 notes predate Yang's 2006 publication. No claim of priority is intended for the standard finite-group identities. This is a revised exposition, not a reproduction of the original source.

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1 Why trace formulas have two sides

1.1 The global picture

Let H be a locally compact unimodular group and $\Gamma \subset H$ a discrete subgroup such that $\Gamma\backslash H$ is compact. Under standard hypotheses on the test function f, right convolution on $L^2(\Gamma\backslash H)$ has a kernel built from the translates of f. Computing its trace by integrating the diagonal of the kernel yields *geometric terms*, grouped by conjugacy classes in Γ. Decomposing the same operator into irreducible representations yields *spectral terms*. In its schematic compact-quotient form,

$$\sum_{\pi} m_{\Gamma}(\pi)\ \operatorname{tr}\pi(f) = \sum_{[\gamma]_{\Gamma}} \operatorname{vol}(\Gamma_{\gamma}\backslash H_{\gamma}) \int_{H_{\gamma}\backslash H} f(x^{-1}\gamma x)\, d\dot{x}. \tag{1}$$

Here H_{γ} and Γ_{γ} are centralizers, and the measures must be chosen compatibly. Arthur's 1989 introductory lecture derives this compact-quotient formula and then considers the case of finite groups, obtaining Frobenius reciprocity as a special case [1, Example 2, p. 14]; see also [15, §1]. A finite group G, with any subgroup Γ, gives a setting in which every sum is finite and none of the analytic convergence difficulties occur.

1.2 The local picture

For a local field F and a connected reductive group $\mathbf{G}$ over F, Arthur starts with the two-sided action of $\mathbf{G}(F) \times \mathbf{G}(F)$ on $L^2(\mathbf{G}(F))$:

$$((a, b) \cdot \phi)(x) = \phi(a^{-1}xb). \tag{2}$$

The trace of an operator obtained from two test functions can again be analyzed geometrically and spectrally. When $\mathbf{G}(F)$ is noncompact, truncation is essential; the general result involves weighted orbital integrals, weighted characters, Levi subgroups, and continuous spectral parameters [12, 13]. For a finite G, however, (2) produces an identity whose two sides involve nothing beyond orbital sums and irreducible characters. This is a model of the *compact-group case* of the local trace formula, not a formula for general reductive p-adic groups. Vign'eras [2, §1.1] uses exactly this two-sided regular representation as the elementary entry point to Arthur's theory, and develops the connection much further.

1.3 Conventions and a map of the results

Every sum over G in this note uses unnormalized counting measure. Thus $\sum_{g\in G} 1 = |G|$. On $\Gamma\backslash G$ we also use counting measure, so a delta function of one coset has norm one. We shall make no tacit substitution between the normalized average $|G|^{-1}\sum_G$ and the unnormalized sum $\sum_G$. The main identities are:

$$\operatorname{tr} R_\Gamma(f) = \sum_{\pi\in\mathrm{Irr}(G)} m_\Gamma(\pi)\ \operatorname{tr}\pi(f) = \sum_{[\gamma]_\Gamma} \frac{\mathcal{O}_G(f,\gamma)}{|\Gamma_\gamma|}, \tag{3}$$

$$\sum_{\pi\in\mathrm{Irr}(G)} \operatorname{tr}\bar\pi(f_1)\ \operatorname{tr}\pi(f_2) = \sum_{[\gamma]_G} \frac{\mathcal{O}_G(f_1,\gamma)\mathcal{O}_G(f_2,\gamma)}{|G_\gamma|}. \tag{4}$$

The orbital sum $\mathcal{O}_G$ is defined below. The centralizers in the two formulas are *different*: Γ_γ in the first, G_γ in the second.

Four complementary precedents. Arthur's finite-group example [1, Example 2] explains the first formula through Frobenius reciprocity. Vignéras [2, §1] starts from the two-sided $G\times G$ action underlying the second formula. Yang [4] develops related trace formulas for coset-space representations, including vector-valued functions, whereas Terras [3, Chs. 22–23] explores finite Selberg-type identities in geometric and spectral settings. These are overlapping but not identical approaches; the purpose of the present note is to connect them using one fixed system of normalization conventions.

2 Finite harmonic analysis: notation and basic facts

2.1 Conjugacy and characters

Write $\mathrm{Irr}(G)$ for the set of isomorphism classes of irreducible finite-dimensional complex representations. We denote the conjugacy class of $\gamma\in G$ by

$$[\gamma]_G = \{x^{-1}\gamma x : x\in G\}, \qquad G_\gamma = C_G(\gamma) = \{x\in G : x\gamma = \gamma x\}.$$

The class has size $|G|/|G_\gamma|$. We distinguish $[\gamma]_G$ from $[\gamma]_\Gamma$ when $\gamma\in\Gamma$. They need not be the same set. Let $\mathcal{C}(G) = \{f : G\to\mathbb{C}\}$ and let $\mathcal{C}(G)^G$ be the subspace of class functions, those constant on each G-conjugacy class. For $\pi\in\mathrm{Irr}(G)$, write $\chi_\pi(g) = \operatorname{tr}(\pi(g))$, and set

$$\pi(f) = \sum_{g\in G} f(g)\pi(g), \qquad \chi_\pi(f) = \operatorname{tr}\pi(f) = \sum_{g\in G} f(g)\chi_\pi(g).$$

The contragredient (equivalently, complex-conjugate unitary) representation is $\bar\pi$, whose character satisfies $\chi_{\bar\pi}(g) = \overline{\chi_\pi(g)} = \chi_\pi(g^{-1})$.

Theorem 2.1 (Character orthogonality and completeness)**.** *The irreducible characters are an orthonormal basis of $\mathcal{C}(G)^G$ for*

$$\langle f,h\rangle_G = \frac{1}{|G|}\sum_{g\in G} f(g)\overline{h(g)}.$$

In particular, $|\operatorname{Irr}(G)| = \#\{G\text{-conjugacy classes}\}$. *For a treatment that derives orthogonality and completeness systematically from the two-sided trace identity, see [2, §§1.1–1.9]; compare the classical character-theoretic approach in [8].*

Remark 2.2. The characters are *not* a basis of all of $\mathcal{C}(G)$ unless G is abelian. Also, the ordinary point masses δ_g have squared norm $1/|G|$ for the normalized inner product above: they are orthogonal, but not normalized. The functions $\sqrt{|G|}\,\delta_g$ are an orthonormal basis of $\mathcal{C}(G)$.

2.2 Counting-measure orbital sums

Definition 2.3. *For $f \in \mathcal{C}(G)$ and $\gamma \in G$, the* orbital sum *is*

$$\mathcal{O}_G(f,\gamma) = \sum_{x \in G} f(x^{-1}\gamma x). \tag{5}$$

It is a class function in γ.

Since each conjugate is attained by $|G_\gamma|$ elements x,

$$\mathcal{O}_G(f,\gamma) = |G_\gamma| \sum_{\eta \in [\gamma]_G} f(\eta). \tag{6}$$

The sum $\sum_{\eta\in[\gamma]_G} f(\eta)$ is the orbital integral with counting measure on $G_\gamma\backslash G$. Consequently the (finite) Weyl integration formula is simply

$$\sum_{g \in G} f(g) = \sum_{[\gamma]_G} \frac{\mathcal{O}_G(f,\gamma)}{|G_\gamma|}. \tag{7}$$

This is just a partition of G into conjugacy classes. It is the elementary precursor of the Weyl-integration rearrangement used in Arthur's compact formula [1, §1] and in finite geometric analogues such as [3, Ch. 22].

Proposition 2.4 (Fourier expansion of an orbital sum)**.** *For every $f \in \mathcal{C}(G)$ and $\gamma \in G$,*

$$\boxed{\mathcal{O}_G(f,\gamma) = \sum_{\pi \in \operatorname{Irr}(G)} \chi_{\bar\pi}(f)\chi_\pi(\gamma).} \tag{8}$$

Proof. Expand the class function $\gamma \mapsto \mathcal{O}_G(f,\gamma)$ in the orthonormal character basis. Its coefficient at χ_π is

$$\begin{aligned}\frac{1}{|G|}\sum_{\gamma\in G} \mathcal{O}_G(f,\gamma)\overline{\chi_\pi(\gamma)} &= \frac{1}{|G|}\sum_{x,\gamma\in G} f(x^{-1}\gamma x)\overline{\chi_\pi(\gamma)} \\ &= \sum_{y\in G} f(y)\overline{\chi_\pi(y)} = \chi_{\bar\pi}(f).\end{aligned}$$

In the second equality we used the conjugation-invariance of χ_π and changed variables $y = x^{-1}\gamma x$. □

Corollary 2.5 (Column orthogonality)**.** *For $\eta, \gamma \in G$,*

$$\sum_{\pi \in \mathrm{Irr}(G)} \overline{\chi_\pi(\eta)}\, \chi_\pi(\gamma) = \begin{cases} |G_\gamma|, & [\eta]_G = [\gamma]_G, \\ 0, & \textit{otherwise.} \end{cases} \tag{9}$$

Proof. Apply (8) with $f = \delta_\eta$. Its left-hand side counts the $x \in G$ for which $x^{-1}\gamma x = \eta$. □

Remark 2.6 (The characteristic function of a class is not a point mass). If $f = \mathbf{1}_{[\eta]_G}$, then $\chi_\pi(f) = |[\eta]_G|\chi_\pi(\eta)$ and $\mathcal{O}_G(f, \gamma) = |G|$ whenever γ is conjugate to η. If instead $f = \delta_\eta$, then $\chi_\pi(f) = \chi_\pi(\eta)$ and the orbital sum in the matching class equals $|G_\eta|$. Confusing these two functions loses a factor of the class size.

3 Two character-table examples

3.1 S_3

The conjugacy classes of S_3 are represented by e, $s = (12)$, and $t = (123)$; their sizes are $1, 3, 2$, and the centralizer orders are $6, 2, 3$. The irreducible representations are the trivial $\mathbf{1}$, the sign representation ε, and the *two-dimensional* standard representation V:

$$V = \{(z_1, z_2, z_3) \in \mathbb{C}^3 : z_1 + z_2 + z_3 = 0\}.$$

Equivalently, $V \cong \mathbb{C}^3/\mathbb{C}(1,1,1)$, but it is *not* $\mathbb{C}^3/V$, which is one-dimensional.

	$[e]$ (size 1)	$[s]$ (size 3)	$[t]$ (size 2)
$\chi_{\mathbf{1}}$	1	1	1
χ_ε	1	-1	1
χ_V	2	0	-1

For example, $1^2 + 1^2 + 2^2 = 6 = |G_e|$, while $1^2 + (-1)^2 + 0^2 = 2 = |G_s|$, in agreement with (9).

3.2 A_5

In A_5 there are five conjugacy classes, with representatives

$$e, \quad a = (12)(34), \quad b = (123), \quad c = (12345), \quad d = c^2 = (13524),$$

and respective sizes $1, 15, 20, 12, 12$. Put

$$\phi_+ = \frac{1 + \sqrt{5}}{2}, \qquad \phi_- = \frac{1 - \sqrt{5}}{2}.$$

The two numbers ϕ_+ and ϕ_- are real algebraic conjugates, *not* complex conjugates. With this notation:

	e	a	b	c	d
class size	1	15	20	12	12
χ_1 (degree 1)	1	1	1	1	1
χ_2 (degree 3)	3	-1	0	ϕ_+	ϕ_-
χ_3 (degree 3)	3	-1	0	ϕ_-	ϕ_+
χ_4 (degree 4)	4	0	1	-1	-1
χ_5 (degree 5)	5	1	-1	0	0

As a useful check, the sum of the squares of the degrees is $1+9+9+16+25 = 60 = |A_5|$. We will revisit the 3-cycle class when proving the local formula.

4 The compact global trace formula for finite groups

4.1 The right-translation representation and its kernel

Fix a subgroup $\Gamma \leq G$. This is the finite compact-quotient construction considered by Arthur [1, Example 2]. For a closely related and more general treatment using vector-valued functions on the coset space, compare Yang [4]. Let

$$\mathcal{C}_\Gamma = \mathbb{C}[\Gamma\backslash G] = \{\varphi : G \to \mathbb{C} : \varphi(\gamma x) = \varphi(x) \text{ for } \gamma \in \Gamma\}.$$

We use the honest right-translation *representation*

$$(R(g)\varphi)(x) = \varphi(xg), \qquad R(g_1)R(g_2) = R(g_1 g_2). \tag{10}$$

(Using $\varphi(xg^{-1})$ instead, with operators composed in the usual order, gives an anti-representation.) For $f \in \mathcal{C}(G)$ define

$$R_\Gamma(f) = \sum_{g\in G} f(g)R(g).$$

Choose counting measure on the coset set. Then

$$(R_\Gamma(f)\varphi)(x) = \sum_{y\in\Gamma\backslash G} K_f(x,y)\varphi(y), \quad K_f(x,y) = \sum_{\gamma\in\Gamma} f(x^{-1}\gamma y). \tag{11}$$

The kernel is well-defined on left cosets in each variable: replacing x or y by a representative of the same coset simply reindexes the sum. In particular,

$$\begin{aligned}\operatorname{tr} R_\Gamma(f) &= \sum_{x\in\Gamma\backslash G} K_f(x,x) = \frac{1}{|\Gamma|}\sum_{x\in G}\sum_{\gamma\in\Gamma} f(x^{-1}\gamma x) \\ &= \frac{1}{|\Gamma|}\sum_{\gamma\in\Gamma} \mathcal{O}_G(f,\gamma).\end{aligned} \tag{12}$$

The factor $1/|\Gamma|$ is the passage from a sum over elements of G to a sum over left cosets. It is *not* a factor $1/|G|$.

4.2 Geometric expansion: two valid ways to group terms

Within Γ, group the summands of (12) by Γ-conjugacy classes. The orbital sum $\mathcal{O}_G(f,\gamma)$ is constant along each such class, and $|[\gamma]_\Gamma| = |\Gamma|/|\Gamma_\gamma|$. Hence

$$\boxed{\operatorname{tr} R_\Gamma(f) = \sum_{[\gamma]_\Gamma} \frac{\mathcal{O}_G(f,\gamma)}{|\Gamma_\gamma|}.} \tag{13}$$

Alternatively, one may group *by G-conjugacy classes*, but then one must count only the elements of each class that lie in Γ:

$$\operatorname{tr} R_\Gamma(f) = \frac{1}{|\Gamma|} \sum_{[\gamma]_G \colon [\gamma]_G \cap \Gamma \neq \varnothing} \Big|[\gamma]_G \cap \Gamma\Big| \mathcal{O}_G(f,\gamma). \tag{14}$$

It is generally *incorrect* to replace $\Big|[\gamma]_G \cap \Gamma\Big|$ by $\Big|[\gamma]_G\Big|$.

Remark 4.1 (Comparison with the compact-quotient coefficient). Define the quotient-orbit sum

$$\mathcal{O}_G^{\mathrm{quot}}(f,\gamma) = \sum_{x \in G_\gamma \backslash G} f(x^{-1}\gamma x) = \frac{\mathcal{O}_G(f,\gamma)}{|G_\gamma|}.$$

Then (13) becomes

$$\sum_{[\gamma]_\Gamma} \frac{|G_\gamma|}{|\Gamma_\gamma|} \mathcal{O}_G^{\mathrm{quot}}(f,\gamma).$$

This is exactly the finite counting-measure counterpart of the coefficient $\operatorname{vol}(\Gamma_\gamma \backslash H_\gamma)$ in (1).

4.3 Spectral expansion and Frobenius reciprocity

Over $\mathbb{C}$ the finite-dimensional representation $\mathcal{C}_\Gamma$ is completely reducible:

$$\mathcal{C}_\Gamma \cong \bigoplus_{\pi \in \mathrm{Irr}(G)} m_\Gamma(\pi)\, \pi. \tag{15}$$

It is the permutation module $\operatorname{Ind}_\Gamma^G \mathbf{1}_\Gamma$, so Frobenius reciprocity gives

$$\boxed{m_\Gamma(\pi) = \dim \pi^\Gamma = \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \chi_\pi(\gamma).} \tag{16}$$

Thus the multiplicity formula is precisely the finite-group Frobenius-reciprocity calculation singled out by Arthur [1, p. 14]. The averaging operator $P_\Gamma = |\Gamma|^{-1} \sum_{\gamma \in \Gamma} \pi(\gamma)$ is the projection onto the Γ-fixed vectors. Taking its trace proves the last equality directly. It follows that

$$\operatorname{tr} R_\Gamma(f) = \sum_{\pi \in \mathrm{Irr}(G)} m_\Gamma(\pi) \chi_\pi(f). \tag{17}$$

The multiplicity $m_\Gamma(\pi)$ is a nonnegative integer; no unexplained dimension or group-order factor belongs in (17).

Theorem 4.2 (Finite compact-global trace formula)**.** *For any finite group G, subgroup $\Gamma \leq G$, and function $f : G \to \mathbb{C}$,*

$$\boxed{\sum_{\pi \in \mathrm{Irr}(G)} m_\Gamma(\pi)\chi_\pi(f) = \sum_{[\gamma]_\Gamma} \frac{\mathcal{O}_G(f,\gamma)}{|\Gamma_\gamma|} = \frac{1}{|\Gamma|}\sum_{\gamma\in\Gamma} \mathcal{O}_G(f,\gamma).} \tag{18}$$

An equivalent Fourier expression is

$$\operatorname{tr} R_\Gamma(f) = \frac{1}{|\Gamma|}\sum_{\pi\in\mathrm{Irr}(G)} \chi_{\bar\pi}(f)\sum_{\gamma\in\Gamma}\chi_\pi(\gamma). \tag{19}$$

Proof. The first equality in (18) equates (17) and (13); the second is (12). Inserting (8) into (12) gives (19). The latter agrees with the first expression because $m_\Gamma(\bar\pi) = m_\Gamma(\pi)$ and $\pi \mapsto \bar\pi$ permutes $\mathrm{Irr}(G)$. □

Example 4.3 (A diagnostic normalization check)*.* Let $G = S_3$, $\Gamma = \{e, (12)\}$, and $f = \delta_e$. Then $R_\Gamma(e)$ acts as the identity on the three-dimensional space $\mathbb{C}[\Gamma\backslash G]$, so its trace is 3. The geometric formula gives

$$\frac{\mathcal{O}_G(\delta_e, e)}{|\Gamma_e|} = \frac{6}{2} = 3.$$

And $\mathrm{Ind}_\Gamma^{S_3}\mathbf{1} \cong \mathbf{1}\oplus V$, hence the spectral side is $1+2=3$. An expression with the coefficient $|G_e|^{-1}$ in place of $|\Gamma_e|^{-1}$ would incorrectly give 1.

4.4 Special cases and correct interpretations

Corollary 4.4 (The regular representation: $\Gamma = \{e\}$)**.** *When $\Gamma = \{e\}$, $m_\Gamma(\pi) = \dim\pi$, and*

$$\sum_{\pi\in\mathrm{Irr}(G)} (\dim\pi)\chi_\pi(f) = |G|\, f(e). \tag{20}$$

Equivalently, $f(e) = |G|^{-1}\sum_\pi(\dim\pi)\chi_\pi(f)$. This is the finite-group Fourier inversion formula at the identity.

Corollary 4.5 (The one-dimensional quotient: $\Gamma = G$)**.** *When $\Gamma = G$, the representation $\mathbb{C}[G\backslash G]$ is trivial:*

$$\sum_{g\in G} f(g) = \sum_{[\gamma]_G} \frac{\mathcal{O}_G(f,\gamma)}{|G_\gamma|}. \tag{21}$$

The right-hand side is the finite Weyl integration formula (7)*. The spectral side has* only the trivial representation*; it is not the full finite-group Plancherel sum.*

Corollary 4.6 (An index-two subgroup)**.** *If $\Gamma \lhd G$ has index two, and $\varepsilon : G \to \{\pm 1\}$ is the nontrivial character of G/Γ, then*

$$\mathbb{C}[\Gamma\backslash G] \cong \mathbf{1}\oplus\varepsilon, \qquad \sum_{[\gamma]_\Gamma}\frac{\mathcal{O}_G(f,\gamma)}{|\Gamma_\gamma|} = \chi_{\mathbf{1}}(f) + \chi_\varepsilon(f) = 2\sum_{\gamma\in\Gamma} f(\gamma).$$

Proof. The two cosets are permuted through the quotient $G/\Gamma \cong C_2$; its permutation representation is $\mathbf{1}\oplus\varepsilon$. □

4.5 A caution about induced representations

The character of $\mathrm{Ind}_\Gamma^G \sigma$, for a representation σ of Γ, is

$$\chi_{\mathrm{Ind}_\Gamma^G \sigma}(g) = \frac{1}{|\Gamma|} \sum_{\substack{x \in G \\ x^{-1}gx \in \Gamma}} \chi_\sigma(x^{-1}gx). \tag{22}$$

The factor $1/|\Gamma|$ is essential. In general the argument "$x^{-1}\Gamma x = \Gamma$" is false: it holds only if Γ is normal (or for x in its normalizer). Thus one cannot conclude that the induced representation has no Γ-fixed vectors just because $\sigma \not\cong \mathbf{1}_\Gamma$.

For example, with $G = S_3$, $\Gamma = \langle (12) \rangle$, and $\sigma = \varepsilon_\Gamma$ the nontrivial character of C_2, one has

$$\mathrm{Ind}_\Gamma^G \sigma \cong \varepsilon_G \oplus V.$$

The restriction $V|_\Gamma \cong \mathbf{1}_\Gamma \oplus \varepsilon_\Gamma$, so $\mathrm{Ind}_\Gamma^G \sigma$ *does* have a Γ-fixed vector. For a normal subgroup $\Gamma \lhd G$, in contrast, the restriction of $\mathrm{Ind}_\Gamma^G \sigma$ is a sum of conjugates of σ, and a nontrivial irreducible σ then gives no Γ-fixed vectors. The appropriate general tool is Mackey's double-coset formula, not the unqualified vanishing claim; see [8, Chapters 7–8]. Yang's coset-space trace formulas [4] provide a complementary framework for extending the scalar permutation-module setting used here.

5 The compact local trace formula for finite groups

5.1 The bilinear identity

Let $f_1, f_2 : G \to \mathbb{C}$, and consider the finite sum

$$I(f_1, f_2) = \sum_{x,y \in G} f_1(x^{-1}yx) f_2(y). \tag{23}$$

This is the two-sided trace computation appearing, in different notation, in Vignéras [2, §1.1]. Using the orbital expansion (8), we obtain

$$\begin{aligned} I(f_1, f_2) &= \sum_{y \in G} \mathcal{O}_G(f_1, y) f_2(y) \\ &= \sum_{\pi \in \mathrm{Irr}(G)} \chi_{\bar{\pi}}(f_1) \chi_\pi(f_2). \end{aligned} \tag{24}$$

On the other hand, partition the y-sum by conjugacy classes. For any class function $A(y)$, $\sum_y A(y) f_2(y) = \sum_{[\gamma]_G} A(\gamma) \mathcal{O}_G(f_2, \gamma)/|G_\gamma|$. Taking $A(y) = \mathcal{O}_G(f_1, y)$ gives

$$I(f_1, f_2) = \sum_{[\gamma]_G} \frac{\mathcal{O}_G(f_1, \gamma) \mathcal{O}_G(f_2, \gamma)}{|G_\gamma|}. \tag{25}$$

Theorem 5.1 (Finite compact-local trace formula)**.** *For arbitrary complex-valued functions f_1, f_2 on G,*

$$\boxed{\sum_{\pi \in \mathrm{Irr}(G)} \chi_{\bar{\pi}}(f_1) \chi_\pi(f_2) = \sum_{[\gamma]_G} \frac{\mathcal{O}_G(f_1, \gamma) \mathcal{O}_G(f_2, \gamma)}{|G_\gamma|}.} \tag{26}$$

Proof. Equate (24) and (25), which both compute (23). □

Remark 5.2 (What Vignéras's treatment adds). Vignéras [2, §1] begins with a field L whose characteristic does not divide $|G|$, rather than only $\mathbb{C}$, and derives Schur's lemma, completeness, and orthogonality from this starting point. We retain $\mathbb{C}$ so that contragredients, complex conjugation, and the positive identity (27) have their familiar Hilbert-space meaning. In characteristic dividing $|G|$, semisimplicity fails in general and the subsequent spectral argument requires a different formulation.

Corollary 5.3 (Positive form; class-function Plancherel)**.** *For every complex-valued f on G,*

$$\boxed{\sum_{\pi\in\mathrm{Irr}(G)} |\chi_\pi(f)|^2 = \sum_{[\gamma]_G} \frac{|\mathcal{O}_G(f,\gamma)|^2}{|G_\gamma|}.} \tag{27}$$

Proof. In (26), put $f_1 = f$ and $f_2 = \bar{f}$. The left-hand side becomes $\sum_\pi |\chi_{\bar\pi}(f)|^2$, which equals $\sum_\pi |\chi_\pi(f)|^2$ after relabeling irreducibles; $\mathcal{O}_G(\bar f,\gamma) = \overline{\mathcal{O}_G(f,\gamma)}$. □

Remark 5.4. This identity is a Parseval/Plancherel statement for the *conjugacy-invariant projection* of f, with the particular unnormalized Fourier and orbital transforms chosen here. It does not assert that character traces determine an arbitrary function on G: they determine its conjugation average.

5.2 A stronger interpretation: the diagonal subgroup

The preceding two identities are not unrelated analogies. There is an exact reduction of the *finite local* identity to the *finite global* identity. The regular $G\times G$ representation itself is the point of departure in Vignéras [2, §1.1]; our additional point here is to obtain it by applying the coset-space theorem to the diagonal subgroup.

Theorem 5.5 (The local identity is a global identity on $G\times G$)**.** *Let $H = G\times G$, let $\Delta G = \{(g,g) : g\in G\} \le H$, and define $F(a,b) = f_1(a)f_2(b)$. Applying Theorem 4.2 to $(H,\Delta G,F)$ gives exactly Theorem 5.1.*

Proof. The map

$$\Delta G\backslash(G\times G) \longrightarrow G, \qquad [(g_1,g_2)] \longmapsto g_1^{-1}g_2$$

is a bijection. Under it, the right action of $(a,b)\in G\times G$ becomes

$$((a,b)\cdot\varphi)(x) = \varphi(a^{-1}xb), \tag{28}$$

Arthur's two-sided regular action. Finite Peter–Weyl theory (or decomposition of the regular bimodule) gives

$$\mathbb{C}[G] \cong \bigoplus_{\pi\in\mathrm{Irr}(G)} \bar\pi\boxtimes\pi \qquad \text{as a } (G\times G)\text{-module.} \tag{29}$$

Each summand occurs once. Its character at (a,b) is $\chi_{\bar\pi}(a)\chi_\pi(b)$. Thus the spectral side for F is $\sum_\pi \chi_{\bar\pi}(f_1)\chi_\pi(f_2)$.

Conjugacy classes of ΔG are indexed by G-conjugacy classes: $[(\gamma,\gamma)]_{\Delta G}$ corresponds to $[\gamma]_G$. Its centralizer in ΔG has order $|G_\gamma|$. Moreover

$$\mathcal{O}_{G\times G}(F,(\gamma,\gamma)) = \sum_{x,y\in G} f_1(x^{-1}\gamma x) f_2(y^{-1}\gamma y) = \mathcal{O}_G(f_1,\gamma)\mathcal{O}_G(f_2,\gamma).$$

The global geometric expansion (13), now for $(H,\Delta G)$, is exactly the right-hand side of (26). This proves the assertion. □

Remark 5.6 (A kernel matching Arthur's construction). The convolution operator for $F = f_1 \otimes f_2$ on $\mathbb{C}[G]$ is

$$(T_F\varphi)(x) = \sum_{a,b\in G} f_1(a) f_2(b) \varphi(a^{-1}xb).$$

Its matrix kernel (with counting measure in the y variable) is

$$K_F(x,y) = \sum_{a\in G} f_1(a) f_2(x^{-1}ay),$$

so

$$\operatorname{tr} T_F = \sum_{x,a\in G} f_1(a) f_2(x^{-1}ax) = I(f_1,f_2).$$

The last equality follows by interchanging the two functions and changing variables in (23). This double-sided kernel is a closer parallel to Arthur's local trace-formula starting point than the bilinear identity viewed solely as character orthogonality; compare [2, §1.1] and [12, §1].

5.3 The character-table matrix is orthogonal after weighting

Choose representatives $\gamma_1,\dots,\gamma_r$ for G-conjugacy classes and irreducibles $\pi_1,\dots,\pi_r$; set $c_j = |G_{\gamma_j}|$. For each j let

$$f_j = |G|^{-1}\mathbf{1}_{[\gamma_j]_G}. \tag{30}$$

Then $\mathcal{O}_G(f_j,\gamma_i) = \delta_{ij}$, by (5). Define

$$a_{ij} = \chi_{\pi_i}(f_j) = \frac{\chi_{\pi_i}(\gamma_j)}{c_j}, \qquad A = (a_{ij})_{1\le i,j\le r}.$$

The local formula for f_i, f_j gives column orthogonality:

$$\boxed{A^*A = \operatorname{diag}(c_1^{-1},\dots,c_r^{-1}).} \tag{31}$$

Equivalently, $A\operatorname{diag}(\sqrt{c_1},\dots,\sqrt{c_r})$ is unitary. *The order of multiplication in* (31) *matters.* The matrix AA^* is generally not diagonal.

Example 5.7 (S_3 revisited). With $\gamma_1 = e$, $\gamma_2 = (12)$, $\gamma_3 = (123)$,

$$A = \begin{pmatrix} 1/6 & 1/2 & 1/3 \\ 1/6 & -1/2 & 1/3 \\ 1/3 & 0 & -1/3 \end{pmatrix}.$$

A direct calculation yields

$$A^*A = \operatorname{diag}(1/6, 1/2, 1/3), \qquad (AA^*)_{12} = -\frac{1}{9} \neq 0.$$

Thus a claim that AA^* is diagonal would be false even for S_3.

Example 5.8 (A_5 and a class indicator). Let $f = \mathbf{1}_{[(123)]_{A_5}}$, the indicator of the class of 20 three-cycles. From the character table,

$$(\chi_1(f), \dots, \chi_5(f)) = (20, 0, 0, 20, -20),$$

so the left side of (27) is $20^2 + 20^2 + (-20)^2 = 1200$. The orbital sum is zero on other classes and is 60 at a three-cycle, since $|A_5| = 60$. Its centralizer has order 3, so the right-hand side is $60^2/3 = 1200$. The equality holds with no hidden measure factors.

6 What these finite identities do and do not model

6.1 The compact-quotient Selberg formula

The finite global formula, Theorem 4.2, reproduces the *form* of the Selberg trace formula for compact quotient: this is the analogy made explicit in Arthur's finite-group example [1, Example 2].

Finite group	Locally compact compact quotient
$\mathbb{C}[\Gamma\backslash G]$	$L^2(\Gamma\backslash H)$
$\chi_\pi(f)$	trace of $\pi(f)$
$m_\Gamma(\pi)$	discrete spectral multiplicity
$\mathcal{O}_G(f,\gamma)/\lvert G_\gamma\rvert$	orbital integral over $H_\gamma\backslash H$
$\lvert G_\gamma\rvert/\lvert\Gamma_\gamma\rvert$	$\operatorname{vol}(\Gamma_\gamma\backslash H_\gamma)$
finite sum on $\Gamma\backslash G$	integral of the kernel on the diagonal

Terras [3, Ch. 22] takes a different, complementary route from trace formulas to finite spectral geometry, including isospectral nonisomorphic examples; Chapter 23 turns to finite upper half planes. Those applications require extra geometric structures and should not be identified with the bare coset-space identity above.

The compact quotient condition should not be confused with compactness of the group itself. For example, a cocompact lattice in a noncompact Lie group yields a compact quotient. Finite groups are a particularly simple case in which both the group and the quotient are compact.

6.2 Why Arthur's global formula is harder

In adelic examples, $H = \mathbf{G}(\mathbb{A}_F)$ (with the appropriate central quotient) and $\Gamma = \mathbf{G}(F)$ for a reductive group over a number field F. The quotient usually is not compact. The kernel need not be integrable along its diagonal, and the spectrum need not be purely discrete. Selberg's and Arthur's theory handles continuous spectrum, parabolic subgroups, Eisenstein series, and truncation; refined formulas feature weighted geometric and spectral terms. None of these difficulties can be detected in the finite counting-measure identity. See [14, 15, 9, 10].

6.3 Why the local analogy is genuinely relevant

Arthur explicitly begins his 1989 article *Towards a local trace formula* with the two-sided $\mathbf{G}(F) \times \mathbf{G}(F)$-action, then works out the compact-group case by a Weyl integration formula and Peter–Weyl decomposition [12, §§1–2]. Theorem 5.5 gives the precise finite-group version of that construction. Vignéras [2, §§1–2] provides an especially useful intermediate presentation from the finite calculation to the p-adic analytic issues. Section 7 analyzes the noncompact transition in detail. When $\mathbf{G}(F)$ is noncompact, the integral of the diagonal kernel may diverge. Arthur's full local trace formula replaces the simple finite sums by a carefully truncated identity whose geometric and spectral expansions include terms indexed by Levi subgroups [13]. In particular, the finite formula is *not* the general Arthur local trace formula with all analytic difficulties suppressed: its lack of proper parabolic/Levi data and continuous spectrum means that major structural features are absent.

7 From finite groups to reductive p-adic groups: why the local trace formula acquires new terms

The finite identity (26) has two ingredients: conjugation orbits on the geometric side and a decomposition of the regular representation on the spectral side. Both ingredients survive for a reductive p-adic group. What fails is the final, seemingly innocent step of integrating the kernel along the diagonal. This section explains that failure and the resulting changes in the formulas. The goal is a conceptual bridge, *not* a proof of Arthur's local trace formula; the latter requires substantial analytic and combinatorial arguments [12, 13]. Vignéras [2, §2] offers a shorter complementary orientation, especially for elliptic elements, weighted orbital integrals, and tempered distributions.

7.1 The setting: Haar measure, Hecke algebra, and admissibility

Let F be a non-Archimedean local field of characteristic zero, with ring of integers $\mathcal{O}_F$, uniformizer ϖ, and residue field of cardinality q. Thus $|\varpi|_F = q^{-1}$. Let $\mathbf{G}$ be a connected reductive algebraic group over F, and write $G = \mathbf{G}(F)$. The group G is locally compact, totally disconnected, and unimodular. Its Haar measure dg is an analogue of counting

measure, but $\mathrm{vol}(G)$ is usually infinite. We use

$$\mathcal{H}(G) = C_c^\infty(G) = \{f : G \longrightarrow \mathbb{C} : \ f \text{ is locally constant and compactly supported}\},$$

where "smooth" in this context means locally constant. For a compact open subgroup $K \subset G$, $\mathbf{1}_K$ belongs to $\mathcal{H}(G)$; we often normalize dg by $\mathrm{vol}(K) = 1$.

The analogue of a finite-dimensional complex representation is now often a *smooth admissible* representation (π, V): for every $v \in V$ the stabilizer of v is open, and for every compact open K the space V^K is finite-dimensional. For such a representation,

$$\pi(f) = \int_G f(g)\pi(g)\, dg$$

is finite-rank when $f \in \mathcal{H}(G)$: choose K fixing f on both sides; then $\pi(f)$ has image in V^K. Its trace $\Theta_\pi(f) = \operatorname{tr} \pi(f)$ is the *distribution character*. Unlike a finite-group character, Θ_π need not be a globally defined ordinary function, although Harish-Chandra's regularity theory represents it by a locally integrable function and by a locally constant function on the regular semisimple locus. For foundational material see [5, 6].

Remark 7.1 (A change of meaning for the diagonal subgroup). For finite G, the identification $\Delta G\backslash(G\times G) \simeq G$ in Theorem 5.5 is an instance of a quotient by a *discrete* subgroup, and the quotient is compact. For a non-discrete p-adic G, the same identification remains valid as an isomorphism of homogeneous spaces, but ΔG is *closed and non-discrete*, while the quotient is usually noncompact. One therefore cannot apply the usual compact-lattice Selberg trace formula to $(G \times G, \Delta G)$ and expect a local trace formula for free. The regular-bimodule analogy persists; the trace-class argument does not.

7.2 The same kernel, but potentially an infinite trace

Let $f_1, f_2 \in \mathcal{H}(G)$, and use the same two-sided representation as in (2):

$$(R(a,b)\phi)(x) = \phi(a^{-1}xb), \qquad \phi \in L^2(G).$$

For $f = f_1 \otimes f_2$, the operator $R(f) = \int_{G\times G} f_1(a)f_2(b)R(a,b)\, da\, db$ has the integral kernel

$$K_f(x,y) = \int_G f_1(xu)f_2(uy)\, du. \tag{32}$$

Indeed, substitute $a = xu$ and then $b = uy$ in the expression for $R(f)\phi(x)$. On the diagonal, the change of variables $u \mapsto x^{-1}u$ yields

$$K_f(x,x) = \int_G f_1(u)f_2(x^{-1}ux)\, du. \tag{33}$$

For a finite group, integrating (33) over x gives exactly $I(f_1, f_2)$ of (23). For a compact group, Haar integration and Peter–Weyl theory similarly give a well-defined trace. For noncompact G, the tempting expression

$$\int_G K_f(x,x)\, dx$$

can diverge. It may still diverge after the obvious noncompactness of the split center has been removed. The following calculation shows this explicitly.

7.3 An exact divergence calculation for $\mathrm{GL}_2(F)$

Proposition 7.2 (Every Cartan shell contributes one unit)**.** *Take $G = \mathrm{GL}_2(F)$, $K = \mathrm{GL}_2(\mathcal{O}_F)$, and $f_1 = f_2 = \mathbf{1}_K$, with $\mathrm{vol}(K) = 1$. Let $Z = F^\times I_2$ be the center and give $Z\backslash G$ Haar measure for which the image $\bar{K}$ of K has volume one. Then $K_f(x,x)$ descends to $Z\backslash G$, but*

$$\int_{Z\backslash G} K_f(x,x)\, d\bar{x} = +\infty. \tag{34}$$

More precisely, with $a_n = \mathrm{diag}(\varpi^n, 1)$, $\bar{a}_n \in Z\backslash G$, and

$$s_0 = 1, \qquad s_n = (q+1)q^{n-1} \quad (n \geq 1),$$

the Cartan double coset $\bar{K}\bar{a}_n\bar{K}$ has volume s_n, while $K_f(x,x) = s_n^{-1}$ everywhere on that coset. Hence the integral over the first $N+1$ double cosets is exactly $N+1$.

Proof. Formula (33) gives

$$K_f(x,x) = \int_G \mathbf{1}_K(u)\mathbf{1}_K(x^{-1}ux)\, du = \mathrm{vol}(K \cap xKx^{-1}). \tag{35}$$

This is unchanged when x is multiplied by a scalar matrix or on either side by an element of K. The Cartan decomposition

$$Z\backslash G = \coprod_{n\geq 0} \bar{K}\bar{a}_n\bar{K}$$

therefore partitions the quotient into shells on which (35) is constant. For $n \geq 1$ the index

$$[K : K \cap a_n K a_n^{-1}] = \#\mathbb{P}^1(\mathcal{O}_F/\varpi^n\mathcal{O}_F) = (q+1)q^{n-1}$$

is s_n; equivalently this is the number of vertices at distance n from a fixed vertex in the $(q+1)$-regular Bruhat–Tits tree of $\mathrm{PGL}_2(F)$. Thus the intersection has volume s_n^{-1}, whereas the associated double coset in $Z\backslash G$ has volume s_n. Each shell contributes $s_n s_n^{-1} = 1$. Summing over all n proves (34). □

Remark 7.3 (What the example proves—and does not prove). This is an actual divergence of the *untruncated diagonal integral*, even modulo the split center $Z = A_G(F)$. It is not a counterexample to Arthur's local trace formula: the latter is built from a truncated kernel and a carefully extracted distribution. The elementary radial cutoff $u_N = \mathbf{1}_{\coprod_{n=0}^N \bar{K}\bar{a}_n\bar{K}}$ yields $\int_{Z\backslash G} K_f(x,x)u_N(x)\, d\bar{x} = N+1$, which illustrates how polynomial growth in a truncation parameter can arise. Arthur's precise truncation is more structured than this radial cutoff; we do *not* identify their constant terms.

7.4 Why the spectral side also changes

Finite Peter–Weyl theory decomposes $\mathbb{C}[G]$ into a finite direct sum of $\bar{\pi} \boxtimes \pi$. In contrast, the unitary regular representation of a reductive p-adic group is described by Harish-Chandra's Plancherel theorem, involving direct *integrals* and a continuous family of tempered representations. One version of Fourier inversion has the schematic form

$$f(e) = \int_{\widehat{G}_{\mathrm{temp}}} \Theta_\pi(f)\, d\mu_{\mathrm{Pl}}(\pi), \qquad f \in \mathcal{H}(G), \tag{36}$$

with the usual interpretation of the measure, characters, and normalizations. Unlike (20), this is generally not a finite character sum.

One source of the continuous parameters is *normalized parabolic induction*. For $G = \mathrm{GL}_2(F)$, let $B = TN$ be the upper triangular Borel subgroup, with

$$T = \left\{ \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} : a, d \in F^\times \right\}, \quad N = \left\{ \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix} : u \in F \right\}.$$

The modulus character is $\delta_B(\mathrm{diag}(a,d)) = |a/d|_F$. For quasicharacters χ_1, χ_2 of $F^\times$, the normalized principal-series representation is

$$\mathrm{Ind}_B^G(\chi_1 \boxtimes \chi_2) = \left\{ \phi : G \to \mathbb{C} : \ \phi(tng) = \delta_B(t)^{1/2} \chi_1(a)\chi_2(d)\phi(g), \quad t = \mathrm{diag}(a,d),\ n \in N \right\}, \tag{37}$$

with the right-translation action and the usual smoothness conditions. Taking $\chi_j(x) = |x|_F^{it_j}$ already gives continuous unitary parameters; reducibility and equivalences must be treated separately. Intertwining operators relate different parabolic realizations. Their normalized forms, and derivatives thereof, enter the weighted spectral distributions of the local trace formula [5, 13].

7.5 Levi subgroups and the geometric indexing

A parabolic subgroup P has a Levi decomposition $P = MN$: M is reductive and N its unipotent radical. Fix a minimal F-Levi subgroup M_0 and denote by $\mathcal{L}(M_0)$ the Levi subgroups containing it. For each M let A_M be the maximal F-split torus in its center. These tori provide the real vector spaces $\mathfrak{a}_M$ used in truncation.

For GL_2 there are only two standard Levi choices:

$$\mathcal{L}(M_0) = \{ M_0 = \mathrm{GL}_1 \times \mathrm{GL}_1,\ G = \mathrm{GL}_2 \}.$$

A regular semisimple element whose characteristic polynomial splits over F is conjugate into M_0; it is elliptic in M_0 but not in G. A regular element with irreducible quadratic characteristic polynomial has a centralizer isomorphic to $E^\times$ for a quadratic field extension E/F, and $E^\times/F^\times$ is compact: it is *elliptic modulo the center* in G. This elementary dichotomy is the rank-one shadow of the Levi-indexed expansion on Arthur's geometric side.

For a regular semisimple γ the ordinary orbital integral is

$$O_\gamma(f) = \int_{G_\gamma \backslash G} f(x^{-1}\gamma x)\, d\dot{x}, \tag{38}$$

with compatible Haar measures. In many conventions a *normalized* orbital integral includes the factor $|D^G(\gamma)|_F^{1/2}$, where

$$D^G(\gamma) = \det(1 - \mathrm{Ad}(\gamma) \mid \mathfrak{g}/\mathfrak{g}_\gamma).$$

This analytic discriminant is not the finite-group centralizer factor $|G_\gamma|$ from (26).

7.6 Truncation, weighted orbital integrals, and weighted characters

Write A_G for the maximal split central torus of G. Arthur introduces a truncation function $u(x,T)$ on $A_G(F)\backslash G$, parametrized by a point T sufficiently far into a positive chamber, and first studies

$$K^T(f) = \int_{A_G(F)\backslash G} K_f(x,x)u(x,T)\,d\bar{x}. \tag{39}$$

In the non-Archimedean case the parameter lies on an appropriate lattice. The truncation is designed to be computable in both the geometric expansion (from Weyl integration) and the spectral expansion (from Plancherel theory).

For a proper Levi M, the geometric expansion involves a weight v_M, built from the relative positions of the parabolic height functions $H_P(x)$ for parabolics with Levi factor M. These heights lie in $\mathfrak{a}_M$ and encode the different ways x approaches infinity along a parabolic direction. An *illustrative*, single-variable weighted orbital integral has shape

$$\int_{G_\gamma\backslash G} f(x^{-1}\gamma x)\,v_M(x)\,d\dot{x}, \tag{40}$$

possibly multiplied by a Weyl discriminant or other measure factors. In Arthur's local formula the functions are on $G\times G$, and the actual weight depends on the pair (x_1,x_2); the final weighted distribution is a *double* orbital integral. For example, in the conventions of [13, §12], for a suitable G-regular $\gamma\in M(F)$ it has the form

$$J_M(\gamma,f) = |D^G(\gamma)|_F \int_{A_M(F)\backslash G}\int_{A_M(F)\backslash G} f_1(x_1^{-1}\gamma x_1)f_2(x_2^{-1}\gamma x_2)v_M(x_1,x_2)\,dx_1\,dx_2. \tag{41}$$

For a concise introduction to the role of these weights, see Vignéras [2, §2.2]. The exact definition of v_M and the domains of validity are part of Arthur's theory; (41) should not be taken as a free-standing definition for arbitrary singular γ. For $M=G$, no proper parabolic direction remains, and the ordinary product-of-orbital-integrals pattern reappears (with the local Weyl-discriminant normalization).

On the spectral side one obtains *weighted characters*: these are no longer just $\Theta_{\bar\pi}(f_1)\Theta_\pi(f_2)$. They involve parabolically induced representations and intertwining-operator data. In particular, taking the derivatives of suitable operator families is a spectral analogue of keeping track of the contributions at infinity. In general the resulting weighted distributions for proper M are not individually conjugation-invariant; Arthur also develops invariant versions.

7.7 The shape of Arthur's identity, and a comparison chart

In a common schematic notation, Arthur's local formula is an equality between sums of the form

$$\underbrace{\sum_{M\in\mathcal{L}(M_0)} c_M \int_{\Gamma_{\mathrm{ell}}(M)} J_M(\gamma,f)\,d\gamma}_{\text{geometric expansion}} = \underbrace{\sum_{M\in\mathcal{L}(M_0)} c_M \int_{\mathscr{T}_{\mathrm{disc}}(M)} a^M(\tau)J_M(\tau,f)\,d\tau}_{\text{spectral expansion}}, \tag{42}$$

where the notation $\mathscr{T}_{\mathrm{disc}}(M)$ is a placeholder for Arthur's discrete spectral parameter data, *not* simply a set of ordinary irreducible representations. The coefficients have the usual Weyl-group/sign pattern

$$c_M = \frac{|W_0^M|}{|W_0^G|}(-1)^{\dim(\mathfrak{a}_M/\mathfrak{a}_G)},$$

with measures and conventions as in Arthur's [13, Theorem 12.2]. This display deliberately suppresses the detailed definitions of $J_M(\tau, f)$, $a^M(\tau)$, parameter spaces, and normalizing factors. It is a *reading guide*, not an independent statement or proof of Theorem 12.2. For GL_2, the two Levi choices have $c_G = 1$ and $c_{M_0} = -\frac{1}{2}$.

Finite-group step	Reductive p-adic replacement
Counting measure on G	Haar measure on a noncompact G
Every function is a test function	Hecke algebra $C_c^\infty(G)$
Finite sum over $\mathrm{Irr}(G)$	Tempered Plancherel direct integral
Ordinary character $\chi_\pi(g)$	Distribution character $\Theta_\pi(f)$
Conjugacy-class sum	Weyl integration over regular semisimple orbits
Kernel diagonal is summable	Kernel diagonal may diverge, even modulo A_G
No proper Levi terms	Levi-indexed geometric and spectral terms
Unweighted orbital products	Weighted orbital distributions $J_M(\gamma, f)$
Product of ordinary traces	Weighted characters $J_M(\tau, f)$
No truncation	Truncation and extraction of a regularized term

7.8 Suggested reading order and the limits of the analogy

A practical route through the literature is to begin with Vignéras [2, §1] beside Theorem 5.5 and Yang [4] beside Theorem 4.2. For the finite-geometric variant, see Terras [3, Chs. 22–23]. Then read [12, §§1–2] alongside the diagonal-subgroup proof, and check the exact GL_2 divergence in Proposition 7.2. Next read [13, §§1–3] for the kernel, Plancherel expansion, and truncation; [13, §§4–6] for the geometric weights; and [13, §§7–12] for the spectral expansion and final identity. The elementary p-adic representation theory needed along the way is developed in [5, 6]; [7] provides the spherical-function viewpoint.

Remark 7.4 (What the finite model predicts, and what it does not). The finite identities correctly predict a *two-sided kernel*, a geometric/spectral comparison, and the importance of Weyl integration and Plancherel theory. They do *not* predict the analytic regularization, the family of parabolic directions, the intertwining-operator terms, or the invariantization procedure. Those are mathematical content, not bookkeeping corrections to the finite formulas.

8 A brief literature review and reading path

Four papers closest to the present approach. The finite-group identities in this note are classical; their significance here is as a teaching device for the geometric/spectral dichotomy. The most directly relevant references are the following.

Arthur's finite-group example (1989). In *The Trace Formula and Hecke Operators* [1, §1, Example 2, pp. 13–14], Arthur specializes the compact-quotient Selberg formula to a finite group G and a subgroup Γ. Choosing a suitable irreducible character as a test function shows that a spectral multiplicity equals $\dim \pi^\Gamma$—the content of Frobenius reciprocity. This is the closest direct precedent for our Theorem 4.2. Arthur also explains why the compact trace formula can be viewed as an extension of finite-group reciprocity to locally compact groups.

Vignéras's local analogue (1991). An Elementary Introduction to the Local Trace Formula of J. Arthur: The Case of Finite Groups [2] begins with the $G \times G$-representation on $L[G]$ for a field L of characteristic not dividing $|G|$. Her §1 derives character orthogonality, complete reducibility, and further structure of the group algebra from a two-sided trace identity. The same regular-bimodule mechanism underlies our Theorem 5.1 and the kernel computation in Theorem 5.5; our proof obtains it from the global coset-space theorem by setting $(H, \Gamma) = (G \times G, \Delta G)$. Vignéras then discusses elliptic elements, weighted orbital integrals, and the actual reductive p-adic local formula in §2, with a final discussion of possible applications and limitations. Her caution is worth emphasizing: a trace identity by itself need not establish the character correspondences one would like.

Terras's finite Selberg analogues (1999). Chapter 22 of *Fourier Analysis on Finite Groups and Applications* [3] develops finite versions of the Selberg trace formula in relation to isospectral spectral geometry. Chapter 23 studies the trace formula on finite upper half planes, and Chapter 24 concerns a tree and Ihara's zeta function. These chapters furnish geometric and spectral applications, beyond the formal identity for $\mathbb{C}[\Gamma\backslash G]$. The distinction matters: an elementary trace identity does not, by itself, imply a graph-theoretic isospectrality theorem.

Yang's coset-space formulas (2006). Yang [4] studies right-regular representations on spaces of vector-valued functions associated with $\Gamma\backslash G$ and states trace formulas building on Vignéras. The scalar permutation module used here is a simpler instance of the coset-space perspective. Because Yang's paper appeared in 2006, it is a later related contribution, not a source for the original 1996–2003 notes.

Finite representation theory. Serre's *Linear Representations of Finite Groups* [8] supplies character orthogonality, induction, and Frobenius reciprocity. In particular, it explains the averaging projection used in (16) and the character-theoretic proof of (8). For a trace-formula-first derivation of these same foundations, compare Vignéras [2, §1].

The compact and noncompact global trace formulas. Arthur's later exposition [15, Part I, §1] sets out the compact quotient and identifies the centralizer-volume coefficient. Gelbart's lectures [14] introduce the general Arthur–Selberg framework, often through GL_2. Arthur's reductive-group papers [9, 10, 11] develop the truncation and invariant distributions needed once the quotient is noncompact.

Background on reductive p-adic groups. Casselman [5] explains smooth and admissible representations, parabolic induction, and intertwining operators; Silberger [6]

provides an analytic treatment. Macdonald [7] is useful for the Cartan decomposition and spherical-function viewpoint used in the $\mathrm{GL}_2(F)$ example of Section 7.

The actual local trace formula. Arthur's *Towards a Local Trace Formula* [12, §§1–2] introduces the $G(F) \times G(F)$ kernel and discusses the compact case; Vignéras [2, §2] offers a particularly accessible parallel introduction. Arthur's *A Local Trace Formula* [13] develops truncation and the Levi-indexed geometric and spectral expansions for the noncompact reductive case. Those weighted terms are the additional substance of the theory, not simply a notational version of our finite sums.

Three guided reading routes. To understand *the global finite analogy*, read Theorem 4.2, Arthur's 1989 Example 2 [1], and Yang [4]. For *the local finite analogy*, read Theorem 5.5 beside Vignéras's §1 [2]. For *geometric applications*, turn to Terras [3, Chs. 22–24]. Only then proceed to Section 7, Vignéras's §2, and Arthur's local papers.

9 Glossary of terms and notation

A_M and $\mathfrak{a}_M$
The maximal F-split torus in the center of a Levi subgroup M and its associated real vector space of logarithmic rational-character data, respectively.

Arthur–Selberg trace formula
A family of identities equating geometric orbital terms with spectral representation-theoretic terms, often after truncation.

Character χ_π
The class function $g \mapsto \operatorname{tr}(\pi(g))$ of a representation π.

Class function
A function constant on conjugacy classes.

Compact quotient
A homogeneous space $\Gamma\backslash H$ which is compact, even if H itself is not compact.

Conjugacy class $\mathrm{Cl}_G(\gamma)$
The set $\{x^{-1}\gamma x : x \in G\}$.

Centralizer G_γ
The subgroup of elements commuting with γ; if $\gamma \in \Gamma$ then $\Gamma_\gamma = G_\gamma \cap \Gamma$.

Contragredient representation $\bar{\pi}$
The dual representation; for a finite group one may choose a unitary model in which $\chi_{\bar{\pi}}(g) = \overline{\chi_\pi(g)}$.

Counting measure
The convention $\int_X f = \sum_{x \in X} f(x)$ for finite X.

Cartan double cosets
In the $\mathrm{GL}_2(F)$ example, the sets $\bar{K}\bar{a}_n\bar{K}$ partition $Z\backslash G$; their measure grows as $(q+1)q^{n-1}$.

Diagonal subgroup ΔG
The subgroup $\{(g,g) : g \in G\}$ of $G \times G$.

Eisenstein series
Functions built from parabolic induction that help describe continuous automorphic spectrum on noncompact arithmetic quotients.

Frobenius reciprocity
The adjunction relating $\mathrm{Hom}_G(\mathrm{Ind}_\Gamma^G \sigma, \pi)$ to $\mathrm{Hom}_\Gamma(\sigma, \mathrm{Res}_\Gamma^G \pi)$.

Geometric side
The trace expansion organized by conjugacy classes and orbital integrals.

Harish-Chandra's Plancherel theorem
Describes the regular representation of a reductive local group by a direct integral of tempered representations and a Plancherel measure.

Induced representation
A representation of G built from one of a subgroup, written $\mathrm{Ind}_\Gamma^G \sigma$.

Invariant vectors π^Γ
Vectors fixed by every element of Γ; their dimension is the multiplicity in (16).

Irreducible representation
A representation with no nonzero proper invariant subspace.

Hecke algebra $\mathcal{H}(G)$
The convolution algebra of compactly supported locally constant functions on a totally disconnected group G.

Levi subgroup
A reductive factor of a parabolic subgroup; Levi subgroups index many terms in Arthur's noncompact global and local expansions.

Local field
A locally compact nondiscrete topological field, such as $\mathbb{R}$, $\mathbb{C}$, or a finite extension of $\mathbb{Q}_p$.

Normalized parabolic induction
Induction from a Levi subgroup of a parabolic, including the factor $\delta_P^{1/2}$ built from the modulus character.

Orbital integral / orbital sum
An integral (or finite sum) of a test function along a conjugacy orbit. Here $\mathcal{O}_G$ is deliberately the *sum over all* $x \in G$; it includes a factor of $|G_\gamma|$ relative to summing once per element of $G_\gamma\backslash G$.

Peter–Weyl decomposition
A decomposition of the regular representation of a compact group into matrix coefficients of irreducibles. For finite G it yields (29).

Schur orthogonality
The orthogonality relations for irreducible matrix coefficients and, in particular, for irreducible characters.

Smooth admissible representation
A smooth representation for which the fixed-vector space under any compact open subgroup is finite-dimensional.

Spectral side
The trace expansion organized by irreducible representations and multiplicities or spectral measure.

Tempered representation
A representation in the Plancherel support, broadly analogous to the part of Fourier analysis that contributes to the regular L^2-spectrum; precise definitions use matrix-coefficient decay.

Test function
A function f used to form a convolution operator; in finite groups it is any element of $\mathbb{C}[G]$.

Truncation
A regularization procedure used when the kernel or its diagonal is not integrable on a noncompact space.

Unipotent radical N
The unipotent normal subgroup in a parabolic decomposition $P = MN$.

Weighted orbital integral / weighted character
A more elaborate geometric or spectral distribution appearing in Arthur's local/global trace formulas for noncompact reductive groups.

Weyl integration formula
A decomposition of integrals into orbital integrals. In the finite setting it is (7).

10 Exercises and optional extensions

Exercise 10.1 (A general S_3 calculation). For $a, b, c \in \mathbb{C}$, set $f = af_1 + bf_2 + cf_3$, using the normalized class indicators in (30) for S_3. Verify directly that

$$\sum_{\pi \in \mathrm{Irr}(S_3)} |\chi_\pi(f)|^2 = \frac{|a|^2}{6} + \frac{|b|^2}{2} + \frac{|c|^2}{3}.$$

Exercise 10.2 (A nonnormal subgroup). For $G = S_3$ and $\Gamma = \langle (12) \rangle$, check the trace formula on the three point masses δ_e, $\delta_{(12)}$, and $\delta_{(123)}$. In the last case, the trace is zero.

Exercise 10.3 (A diagonal-global computation). Apply Theorem 4.2 directly to $(G \times G, \Delta G)$ with $f_1 = \delta_u$ and $f_2 = \delta_v$. Recover column orthogonality (9).

Exercise 10.4 (The GL_2 shell count). For a finite ring $R = \mathcal{O}_F/\varpi^n\mathcal{O}_F$, verify that $\#\mathbb{P}^1(R) = q^n + q^{n-1}$. Deduce $[K : K \cap a_n K a_n^{-1}] = (q+1)q^{n-1}$.

Exercise 10.5 (Two kinds of elliptic elements). In $\mathrm{GL}_2(F)$, compare the centralizers of $\mathrm{diag}(a, d)$, $a \neq d$, and a regular matrix with irreducible quadratic characteristic polynomial. Explain why one class is elliptic only in the diagonal Levi and the other is elliptic modulo the center in G.

Exercise 10.6 (Arthur's Frobenius-reciprocity example). Fix $\pi \in \mathrm{Irr}(G)$ and set $f_\pi(g) = \chi_\pi(g^{-1})$. Use Theorem 4.2 and the orthogonality of irreducible characters to isolate $m_\Gamma(\pi)$. Recover $m_\Gamma(\pi) = \dim \pi^\Gamma$ and compare the calculation with [1, Example 2].

Exercise 10.7 (Reading Vignéras and Terras). Compare Vignéras's characteristic-sum notation [2, §1.1] with $\mathcal{O}_G(f, \gamma)$ in (5). Then consult Terras [3, Ch. 22]: what finite geometric or spectral data enter beyond the permutation module $\mathbb{C}[\Gamma\backslash G]$?

A A short exact sequence and group cohomology

The conjugation action of G on $\mathcal{C}(G)$ is $(h \cdot f)(x) = f(h^{-1}xh)$. Its averaging projection is

$$(Pf)(x) = \frac{1}{|G|} \sum_{h \in G} f(h^{-1}xh) = \frac{1}{|G|} \mathcal{O}_G(f, x).$$

It satisfies $P^2 = P$ and has image $\mathcal{C}(G)^G$, the class functions. Moreover,

$$\ker P = \mathrm{span}_{\mathbb{C}}\{f - h \cdot f : f \in \mathcal{C}(G),\ h \in G\} =: I(G).$$

Consequently

$$0 \longrightarrow I(G) \longrightarrow \mathcal{C}(G) \xrightarrow{P} \mathcal{C}(G)^G \longrightarrow 0$$

is a split exact sequence of complex G-modules. The word "span" is necessary: the *set* of all single differences $f - h \cdot f$ need not itself be a linear subspace. Since G is finite and the coefficient field has characteristic zero, averaging makes the invariants functor exact. In particular, for every complex G-module M and every $n > 0$,

$$H^n(G, M) = 0.$$

Thus all higher cohomology groups of $I(G)$, $\mathcal{C}(G)$, and $\mathcal{C}(G)^G$ vanish. The nontrivial information here is in degree zero: $\mathcal{C}(G)^G$ is the class-function space and $I(G)^G = 0$.

B Summary of substantive corrections to the original notes

1. Corrected S_3's standard representation from the wrong quotient $\mathbb{C}^3/\{z_1+z_2+z_3=0\}$ to the two-dimensional subspace $\{z_1+z_2+z_3=0\}$ (or quotient by the diagonal line).
2. Distinguished point masses from class indicators in the character-table argument; the latter have a class-size factor.
3. Distinguished the normalized character inner product from unnormalized point masses, and noted that characters span class functions rather than all functions on G.
4. Replaced the inverse-argument right translation, which was an anti-representation under ordinary composition, by the correct representation $R(g)\varphi(x)=\varphi(xg)$.
5. Recomputed the kernel with counting measure on $\Gamma\backslash G$; fixed the factors $|\Gamma|$ and $|G|$.
6. Corrected the geometric side: sum over Γ-conjugacy classes with $1/|\Gamma_\gamma|$, or sum over G-classes with $|[\gamma]_G\cap\Gamma|/|\Gamma|$.
7. Replaced the faulty extraction of spectral multiplicities by the projection formula $m_\Gamma(\pi)=\dim\pi^\Gamma$.
8. Supplied the missing $1/|\Gamma|$ in the induced-character formula and removed the unjustified assumption that every subgroup is stable under conjugation.
9. Replaced the index-two argument by the direct identity $\mathrm{Ind}_\Gamma^G\mathbf{1}=\mathbf{1}\oplus\varepsilon$.
10. Corrected the A_5 notation: $(1\pm\sqrt{5})/2$ are real algebraic conjugates, not complex conjugates.
11. Corrected the character-table matrix identity to $A^*A=\mathrm{diag}(|G_{\gamma_j}|^{-1})$, not AA^*.
12. Kept the valid bilinear local identity, strengthened it to complex functions with a positive Hermitian form, and identified it as the global trace formula for $(G\times G,\Delta G)$.
13. Distinguished the compact local analogue from the full Arthur local trace formula, which requires truncation and weighted terms in noncompact situations.
14. In the subsequent expansion, showed explicitly for $\mathrm{GL}_2(F)$ that the untruncated local diagonal integral diverges even modulo the center and detailed the role of Levi groups, parabolic induction, and weighted distributions.

C Concluding perspective

The finite trace formulas rest on two familiar operations: computing the trace of a convolution operator on its diagonal, and computing the trace after decomposing a representation into

irreducibles. For $\mathbb{C}[\Gamma\backslash G]$ this gives the compact-global identity; for the two-sided action of $G \times G$ on $\mathbb{C}[G]$ it gives the compact-local identity. Their common structure is especially transparent because the second is literally the first applied to the diagonal subgroup. The main conceptual leap in Selberg's and Arthur's work is to make related comparisons meaningful for noncompact groups and quotients, where continuous spectrum, parabolic induction, and truncation all become central. The precise $\mathrm{GL}_2(F)$ calculation in Section 7 shows why the passage to the local trace formula is not merely a change from finite sums to integrals.

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